EI-SCORE

Explainable-**I**nterpretable
Simple **C**ustomized **O**verall **R**anking **E**ngine

User Guide

Version 0.9.0 | 2026

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Contents

1. Introduction

EI-SCORE is an interactive, browser-based application for building Explainable and Interpretable Composite Indicators using decision rules. It uses the Dominance-based Rough Set Approach (DRSA) (Greco et al., 2001) as described by Corrente et al. (2026)

The tool is designed for researchers, analysts, and decision-makers who need to rank or classify a set of units (countries, companies, students, patients, etc.) evaluated on multiple criteria, and who require transparent, human-readable justifications for the classification/scoring assigned.

1.1 Key features

- **Rule-based representation of the composite indicator**: "If-then" decision rules link a conjunction of elementary conditions on selected individual indicators (criteria) to a unit's class or score.
- **Classification problems** - assigning alternatives to ordered classes (e.g., low, medium, high).
- **Scoring problems** - evaluating and ranking alternatives by a real-valued score.
- **Rule induction** - inducing at-least ($\mathcal{R}^{\geqslant}$) and at-most ($\mathcal{R}^{\leqslant}$) decision rules from classified or scored reference units.
- **Full pipeline** - inducing a maximal and a minimal set of rules able to classify or score all units without any contradiction.
- **New unit assignment** - select a maximal and minimal set of rules able to classify or score with no contradiction a new set of units together with the previously considered sets of units.
- **Rule reuse** - visualize saved rules and apply to every dataset matching criteria.
- **Missing value support** - automatic detection and handling of missing data in the description of units.

References

Salvatore Corrente, Salvatore Greco, Roman Słowiński, and Silvano Zappalà. An explainable and interpretable composite indicator based on decision rules. *Omega*, 142:103513, 2026. doi: 10.1016/j.omega.2026.103513.

Salvatore Greco, Benedetto Matarazzo, and Roman Słowiński. Rough sets theory for multicriteria decision analysis. *European Journal of Operational Research*, 129(1):1–47, 2001. doi: 10.1016/S0377-2217(00)00167-3.

2. Installation and Launch

EI-SCORE can be used in two ways: as a locally installed application or via a web browser without any installation.

2.1 Option A - Web access (no installation required)

The application is publicly available online. Simply open the following URL in your browser:

`https://ei-score.streamlit.app`

No Python or package installation is required.

2.2 Option B - Local installation

Requirements:

- Python 3.14.3 or later.
- `numpy==2.4.2`, `pandas==3.0.1`, `scipy==1.17.0`, `streamlit==1.58.0` `openpyxl==3.1.5`.

Step 1 - Clone the repository:

```
git clone https://github.com/silvanozap/Explainable-composite-
    indicator.git
cd Explainable-composite-indicator
```

Step 2 - Install dependencies:

```
pip install -r requirements.txt
```

Step 3 - Launch the application:

```
streamlit run drsa/interface/app.py
```

The application opens automatically at `http://localhost:8501`.

3. Input File Formats

3.1 Units file - load units in EI-SCORE

The data file must be a EXCEL, CSV or TXT file with the following structure:

```
Name,criterion_1,criterion_2,...,criterion_k,assignment
A1,4,3,2,1
A2,5,4,3,2
A3,7,6,5,
A4,6,5,4,
```

- **First column**: if the first column contains non-numeric values, it is automatically interpreted as unit names.
- **Criteria columns**: quantitative values or qualitative values. Any column separators (comma, semicolon, tab, space) are supported and selectable in the interface.
- **Last column**: class/score label with increasing preference direction.
  - Non-empty value →. **reference unit** (used for rule induction)

  - Empty / NaN → **non-reference unit**.

- **Missing values**: empty cells in criteria columns are detected and handled automatically.

It is possible to use a comma (`,`), semicolon (`;`), tab (`Tab`) or space as a separator. Files must be encoded as `UTF-8`. When uploading an EXCEL file, it must respect the same structure, with the dataset already organized into columns.

Tip

Sample data files covering all use cases are available in the `examples/` folder of the repository, and can also be downloaded directly from the DATA & SETUP tab.

Warning

The class/score label has always an increasing direction of preference

The units file can be uploaded in the SIDEBAR, INCREMENTAL LEARNING, APPLY RULES tabs (Sections 4, 7 and 8).

3.2 Rules file (classification) - Load previous classification rules induced

Classification rules exported from the app or created manually follow this format:

```
#directions,increasing,decreasing,...
#mode,class
type,assignment,criterion_1,criterion_2,...
at-least,2,0.762,,
at-most,1,,2.71,
```

- **`#directions`**: preference direction of each criterion (`increasing` or `decreasing`).
- **`#mode`**: `class` for classification problems.
- **Data rows**: each row is a rule with `type` (`at-least` or `at-most`), `class` (assigned class), and threshold values for each criterion (empty if the criterion is not in the rule condition).

For example, the following row:

```
at-least, 2, 0.2762, ,
```

corresponds to an at-least rule classifying units to at-least class 2 if its performance on criterion_1 is at least 0.762.
The rules file can be uploaded in the APPLY RULES tab (Sections 8).

3.3 Rules file (scoring) - Load previous scoring rules induced

For scoring problems, the `#mode` line changes and class values represent score values:

```
#directions,increasing,increasing,...
#mode,score
type,assignment,criterion_1,criterion_2,...
at-least,16.67,0.762,,
at-most,0,,2.71,
```

For example, the following row:

```
at-most,0,,2.71,
```

corresponds to an at-most rule giving to a unit at-most score 0 if its performance on criterion_2 is at most 2.71.
The rules file can be uploaded in the APPLY RULES tab (Sections 8).

3.4 Rules file (qualitative criteria) - Load previous rules induced with qualitative criteria

For classification problems where some criteria are qualitative, the rules file contains `#mapping` for each qualitative criterion:

```
#directions,increasing,decreasing,...
#mode,class
#mapping,criterion_1,sufficient:1,good:2
#mapping,criterion_2,small:1,big:2
#mapping,assignment,down:1,top:2
type,assignment,criterion_1,criterion_2,...
at-least,2,1,,
at-most,1,,3,
```

For example, the following row:

```
at-least,2,1,,
```

corresponds to an at-least rule giving to a unit at-least class “top” if its performance on criterion_2 is not worse than “sufficient”.
The rules file can be uploaded in the APPLY RULES tab (Sections 8).

Tip

Sample rules files for both classification and scoring can be downloaded directly from APPLY RULES tab of the app.

4. Tab 1 - DATA & SETUP

The DATA & SETUP tab is the starting point of every analysis. It allows the user to load data, configure the problem, and set all parameters of the analysis.

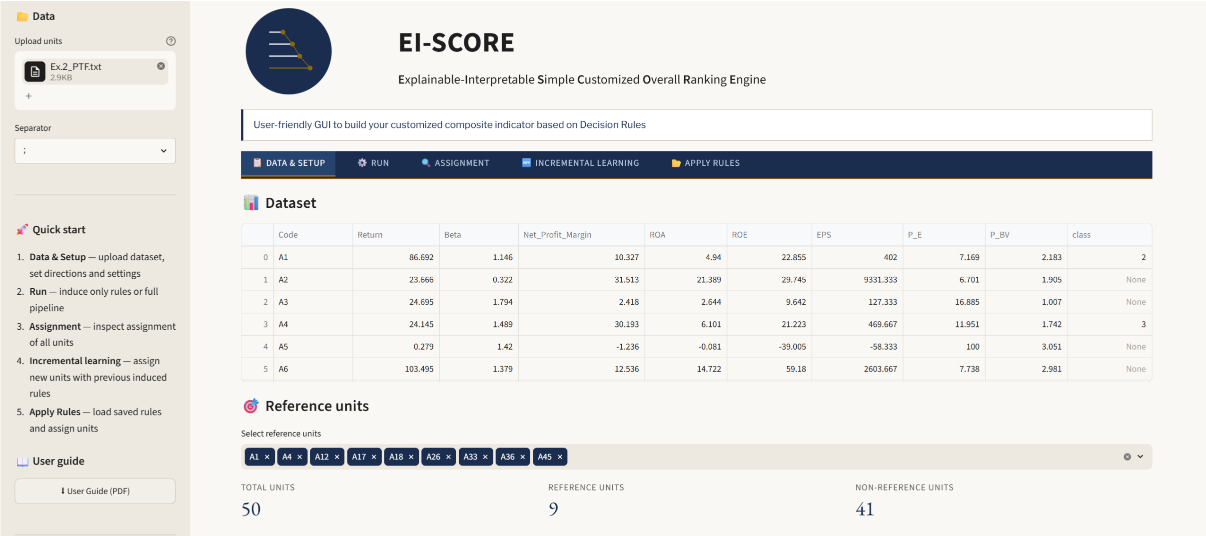

Figure 1: Tab DATA & SETUP - Overview

4.1 Upload data

Click the **Upload EXCEL, CSV or TXT** button in the sidebar to load your data file. Select the appropriate column separator (comma, semicolon, tab, or space) from the dropdown menu. Once uploaded, the dataset is displayed in the main area.

4.2 Reference units

Reference units are the ones whose class/score assignments are known and are used for rule induction. By default, all units with a non-empty class/score label (last column of the loaded dataset) are considered reference units. You can manually select a subset of reference units using the multiselect widget.

Tip

In the Run tab, non-reference units do not participate in *rule induction only* but participate in *full pipeline*.

4.3 Preference directions of criteria

For each criterion, select the preference direction:

- **Increasing (↑) (default)** - higher values are preferred.
- **Decreasing (↓)** - lower values are preferred.

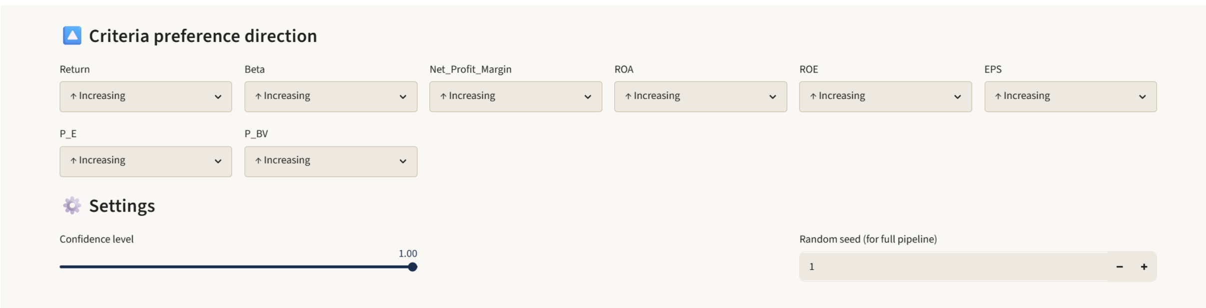

Figure 2: Tab DATA & SETUP - Preference direction of criteria selection

4.4 Qualitative criteria

Qualitative criteria are automatically recognized by the app.

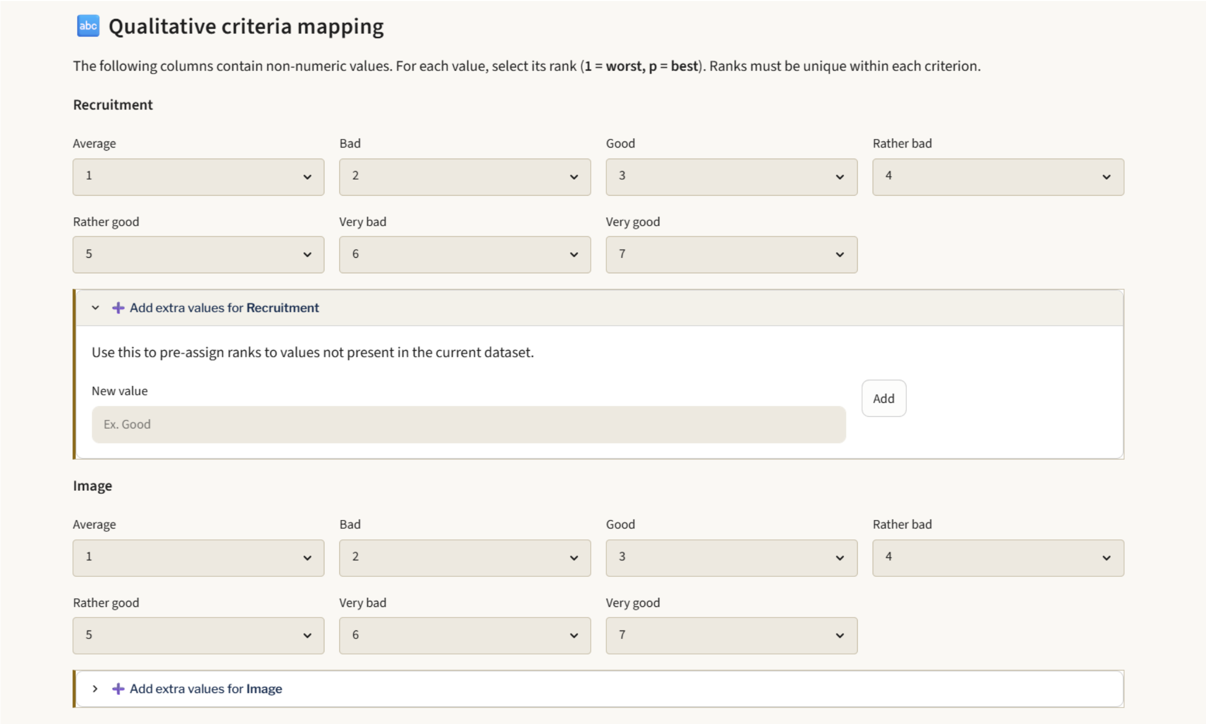

Figure 3: Tab DATA & SETUP - Mapping of qualitative criteria levels onto a numerical scale

EI-SCORE automatically identifies the qualitative values present in each criterion. For each of them, the user must map the ordered performance levels onto a numerical scale. If the loaded dataset does not contain all the values that a criterion may assume, the user can add the missing ones and assign them a rank consistent with the others. This is particularly important when new units added in the Incremental Learning tab may carry evaluations not present in the original dataset.

Tip

Alternatively, you may manually convert qualitative performance levels into numerical values before loading them in the app.

Warning

Ensure the file is properly formatted since any non-identical entries will be recognized as different values.

4.5 Settings

- **Min confidence (c) (default: 1)**: minimum confidence threshold for rule induction. Reducing this value below 1 allows the induction of approximate rules.
- **Random seed (default: 1)**: controls the random number generator used in the greedy selection step in case of running *full pipeline* in the RUN tab.

Warning

Notice that the loaded class/score is by default recognized as an increasing direction of preference.

5. Tab 2 - RUN

Tab RUN is where *rule induction* and the *full pipeline* are executed.

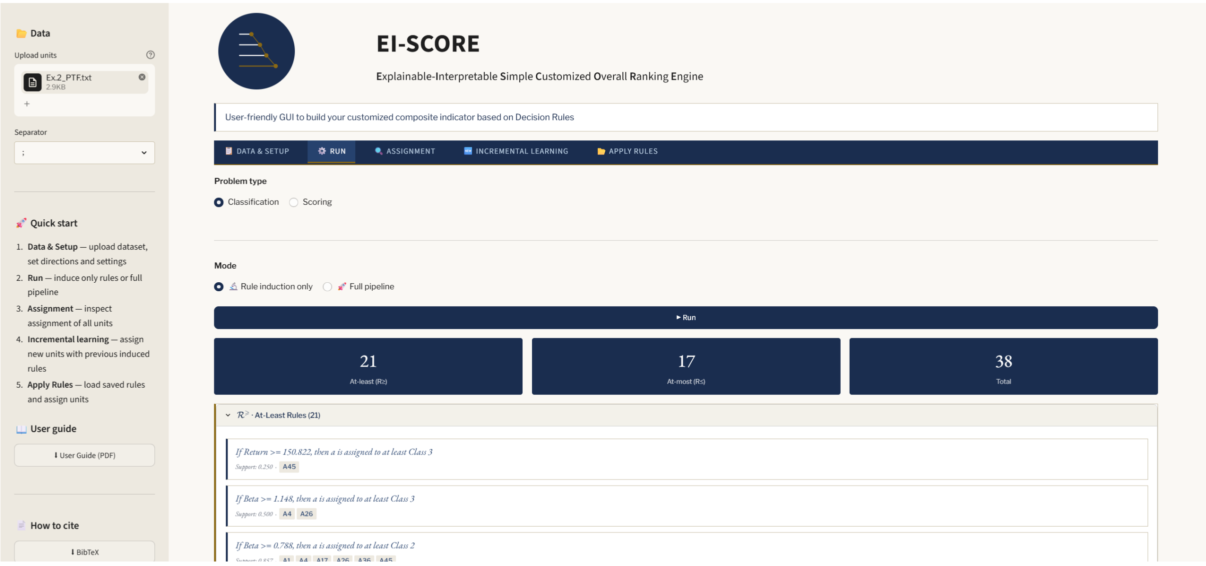

Figure 4: Tab RUN - Rule induction only

5.1 Problem type

Select the problem type using the **Problem type** radio button:

- **Classification (default)** - the last column represents a class code (integer number).
- **Scoring** - the last column represents scores (continuous number).

5.2 Rule induction only

Select **Rule induction only**: induction of at-least ($\mathcal{R}^{\geqslant}$) and at-most ($\mathcal{R}^{\leqslant}$) decision rules from the set of reference units.

The induced rules are displayed in expandable section, showing:

- Rule text with condition and decision part.
- Support value (fraction of reference units covered).
- Matching units (units matching the condition part of the rule).

Warning

The induced rules may provide contradictory classification for non-reference units in the next ASSIGNMENT tab (Section 6).

5.2.1 Minimal rule set

When all units in the dataset are selected as reference units, the **Find minimal set** button appears below the induced rules. Clicking on the button solves a Mixed Integer Linear Programming (MILP) problem finding the smallest subset of the induced rules that preserves the same classification of all reference units.

The minimal rule set is displayed below the induced rules, together with a download button.

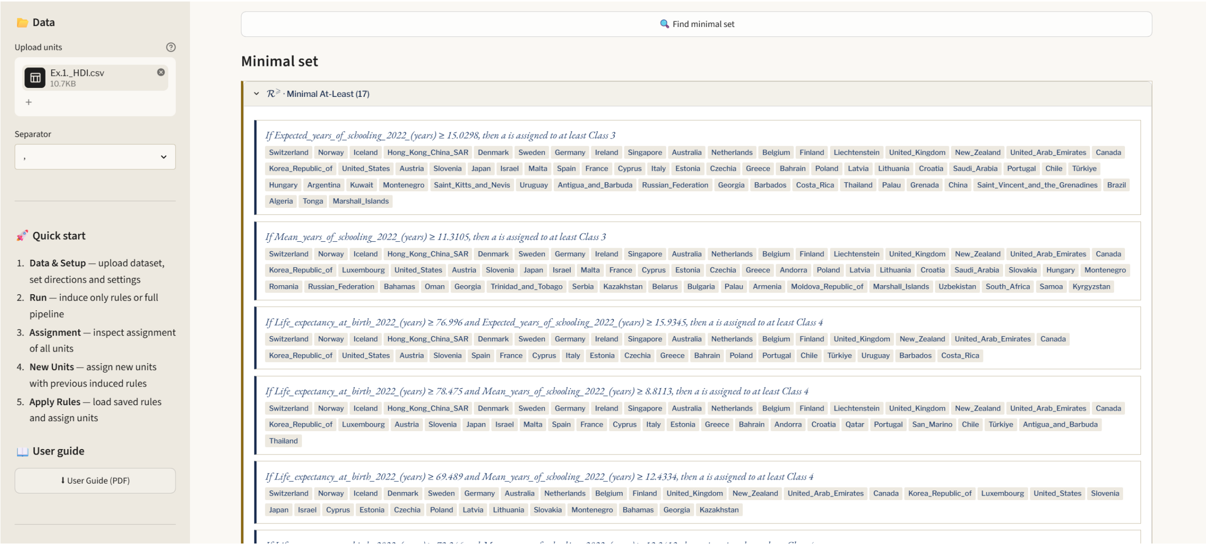

Figure 5: Tab RUN - Rule induction only finds a minimal set of rules

5.3 Full pipeline

Select **Full pipeline** to run the rule induction avoiding any contradictory assignment. The computation is composed of the following steps:

- **Step 1**: Rule induction from classified or scored reference units.
- **Step 2**: Greedy selection of rules that avoid contradictory assignment of all units.
- **Step 3**: Assignment of all units to classes or scores using the selected rules.
- **Step 4**: Rule induction from all units classified or scored to create the maximal set of rules.
- **Step 5**: MILP-based minimization to extract a minimal set of rules reproducing the assignments from Step 3.

A progress bar shows the pipeline execution. Upon completion, a summary table shows the number of rules at each step and maximal and minimal sets of rules.

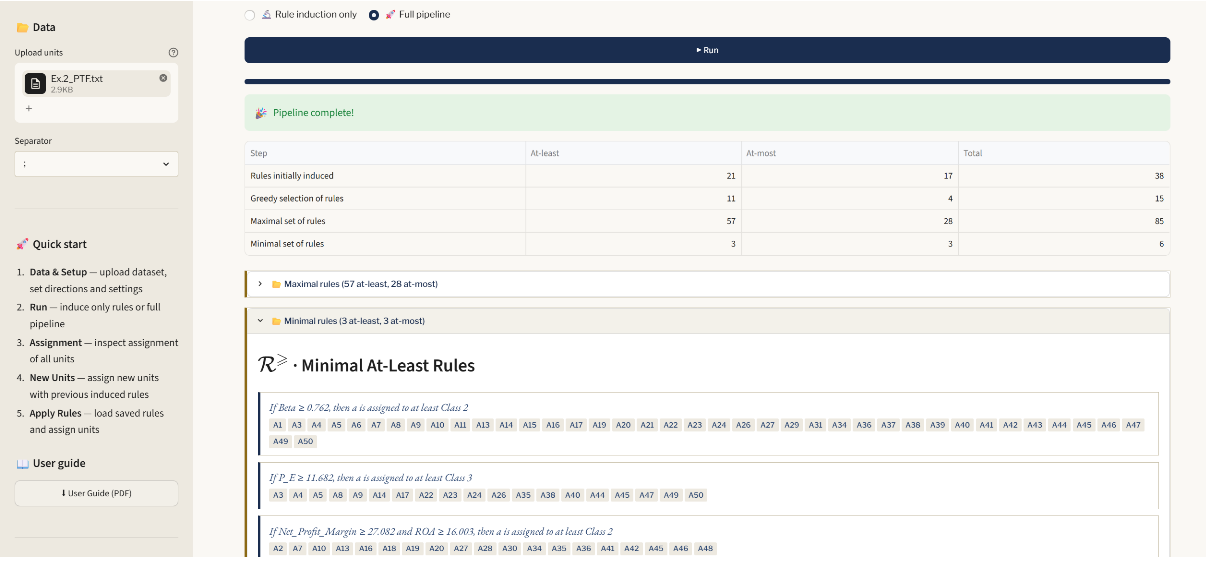

Figure 6: Tab RUN - Full pipeline summary

5.4 Exporting rules

After running, the following CSV files can be downloaded:

- **Maximal rules CSV**: the full set of rules.
- **Minimal rules CSV**: a minimal set of rules.

Tip

Downloaded CSV can be used to load rules in the APPLY RULES tab.

6. Tab 3 - ASSIGNMENT

The ASSIGNMENT tab displays the classification decision of all units by the last set of induced rules.

6.1 Assignment table

For each unit, the table shows:

- **Minimal assignment**: best class/score supported by at-least rules (lower bound).
- **Maximal assignment**: worst class/score supported by at-most rules (upper bound).
- **Assignment**: assignment result:
  - Minimal assignment equals to Maximal assignment: unit assigned to a single class/score.
  - Minimal assignment is lower than Maximal assignment: unit assigned to a range of classes/scores (uncertain).
  - Minimal assignment is higher than Maximal assignment: contradictory class/score assignment.

  The assignment table shows the qualitative value for any qualitative criterion.
- **Status**: Contradictory assignment check:

- OK: Minimal assignment is not higher than Maximal assignment.
- Contradictory: Minimal assignment higher than Maximal assignment.

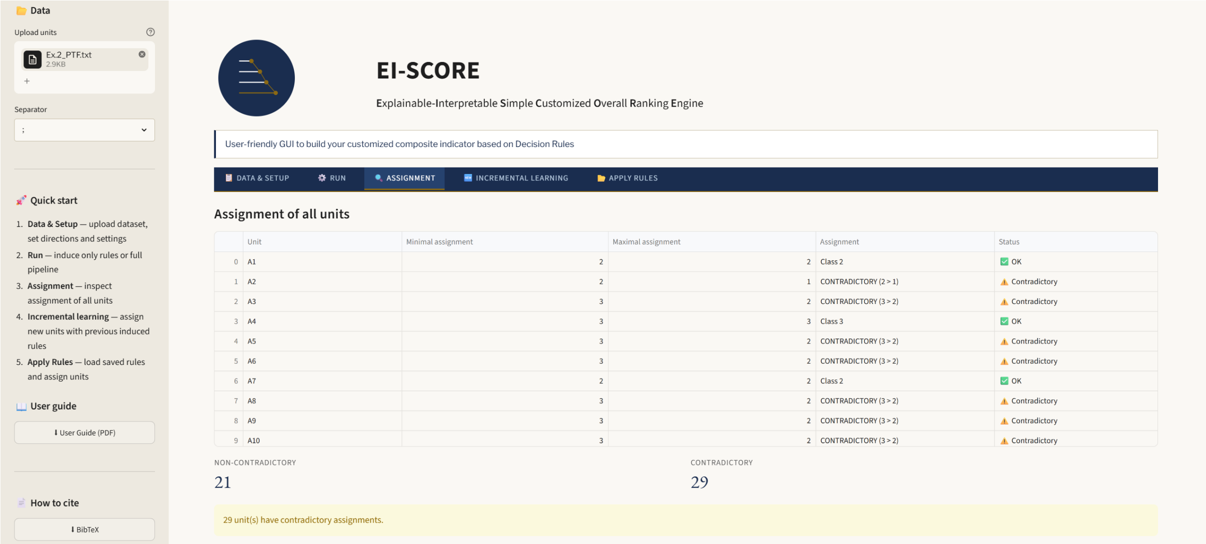

Figure 7: Tab ASSIGNMENT - Assignment table

> **Tip**
>
> Contradictory classification may appear only if you run *Rule induction only* in the RUN tab).

6.2 Unit-by-unit explanation

Select a unit from the drop-down menu to see a detailed **explanation** of its classification/scoring: which rules were matched, and what Minimal and Maximal assignments were determined.

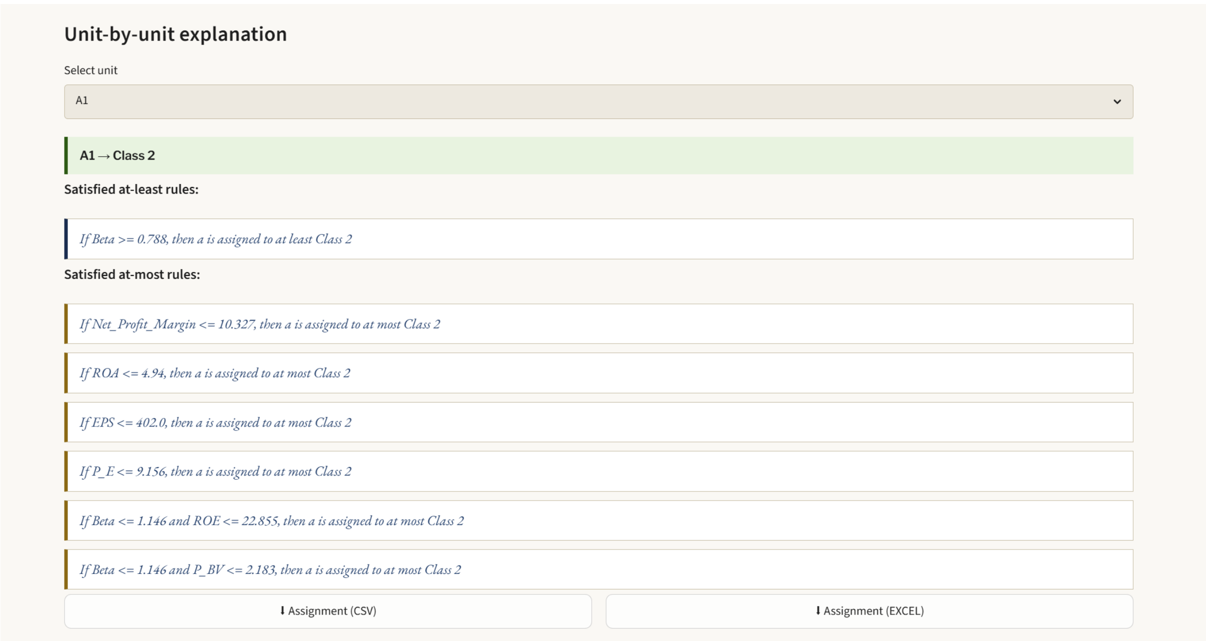

Figure 8: Tab ASSIGNMENT - Unit-by-unit explanation

A downloadable CSV or EXCEL file is also available.

7. Tab 4 - INCREMENTAL LEARNING

The INCREMENTAL LEARNING tab allows classifying or scoring new units that were not part of the original dataset.

7.1 Uploading new units

Upload an EXCEL, CSV or TXT file containing the new units. Once uploaded, the table of new units appears.

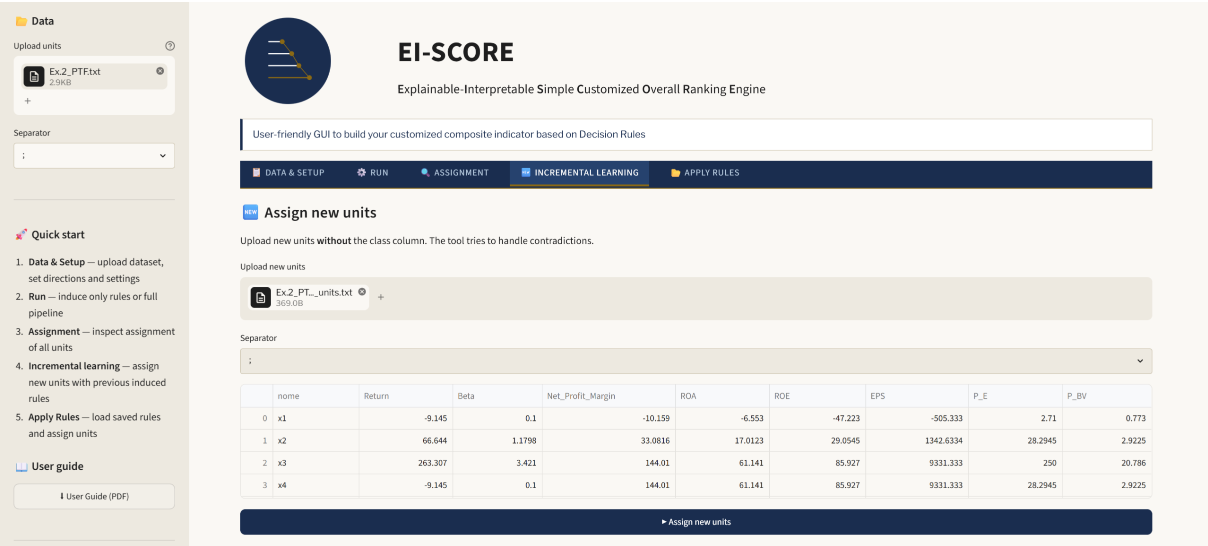

Figure 9: Tab INCREMENTAL LEARNING - New units loaded

Warning

The *full pipeline* must be run in the RUN tab before new units can be assigned to class or score.

Warning

The file must include the same criteria columns as the original dataset.

7.2 Assignment results

By clicking the button *Assign new units*, maximal and minimum sets of rule are found. Both of them are a subset of the maximal rules found in the RUN tab. The results table shows all original units together with the new units (highlighted in yellow), displaying:

- **Minimal assignment (previous)**: initial lower bound classification/score using the maximal set of rules found in the RUN tab.
- **Minimal assignment (previous)**: initial lower bound classification/score using the maximal set of rules found in the RUN tab.
- **Minimal assignment (new)**: final lower bound classification/score using the set of maximal rules selected.

- **Maximal assignment (new)**: final upper bound classification/score using the set of maximal rules selected.
- **Assignment**: final decision assignment.
- **Changed**: whether the assignment changed from the initial due to the inclusion of new units and the selection of rules.

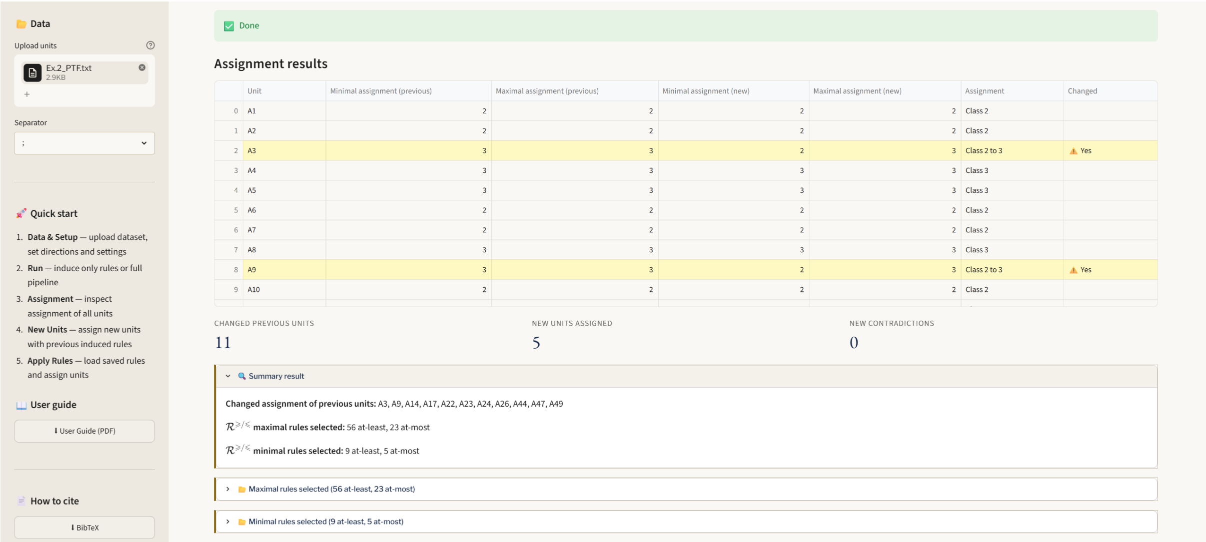

Figure 10: Tab INCREMENTAL LEARNING - New units assignment results

A compact summary result is available showing also the maximal set of selected rules and a minimal set of selected rules able to provide the same classification.

7.3 Unit-by-unit explanation

The unit-by-unit explanation is available for all units (original and new), using the minimal set of rules. It is possible to download complete units assignment, maximal and minimal set of selected rules.

8. Tab 5 - APPLY RULES

The APPLY RULES tab allows loading a previously saved rules file and applying it to compatible units (with the same criteria), without needing to re-run rule induction. This option is useful for an incremental learning of rules.

> **Tip**
>
> This tab is always accessible, even without loading a data file in the sidebar.

8.1 Loading rules

Upload a rules CSV file (in the format described in Section 3). The application automatically reads the `#directions` (increasing or decreasing), `#mode` (class or score) and `#mapping` (rank of qualitative entries for each criterion) lines to reconstruct the criterion directions, problem type and if any qualitative criteria (see Section 3). Upon loading, the application displays:

- Number of at-least and at-most rules.
- Criteria names and directions.

- List of uploaded at-least and at-most rules.

8.2 Visualizing rules

The uploaded rules are displayed and, if units have been loaded, the matching units are highlighted.

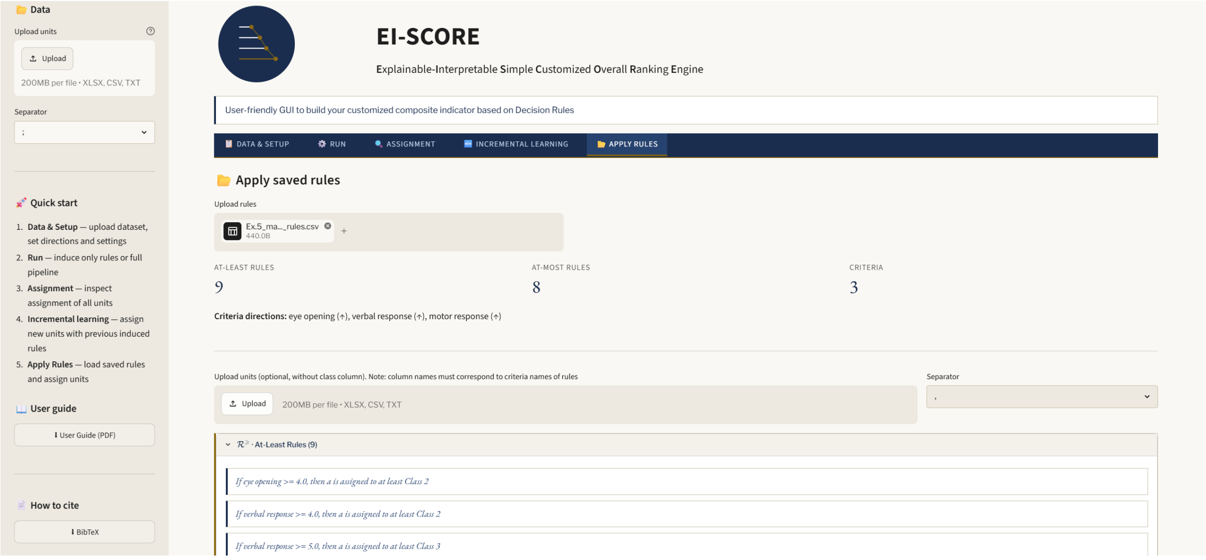

Figure 11: Tab APPLY RULES - Uploaded rules visualization

8.3 Loading units and view assignment

Optionally, you can upload a file of units to assign them a class or score using the loaded rules. The assignment table and unit-by-unit explanation are displayed.

> **Warning**
>
> The column names in the units file must exactly match the criterion names in the rules file.

> **Warning**
>
> In case of qualitative criteria, all values must be mapped in the corresponding `#mapping` line.

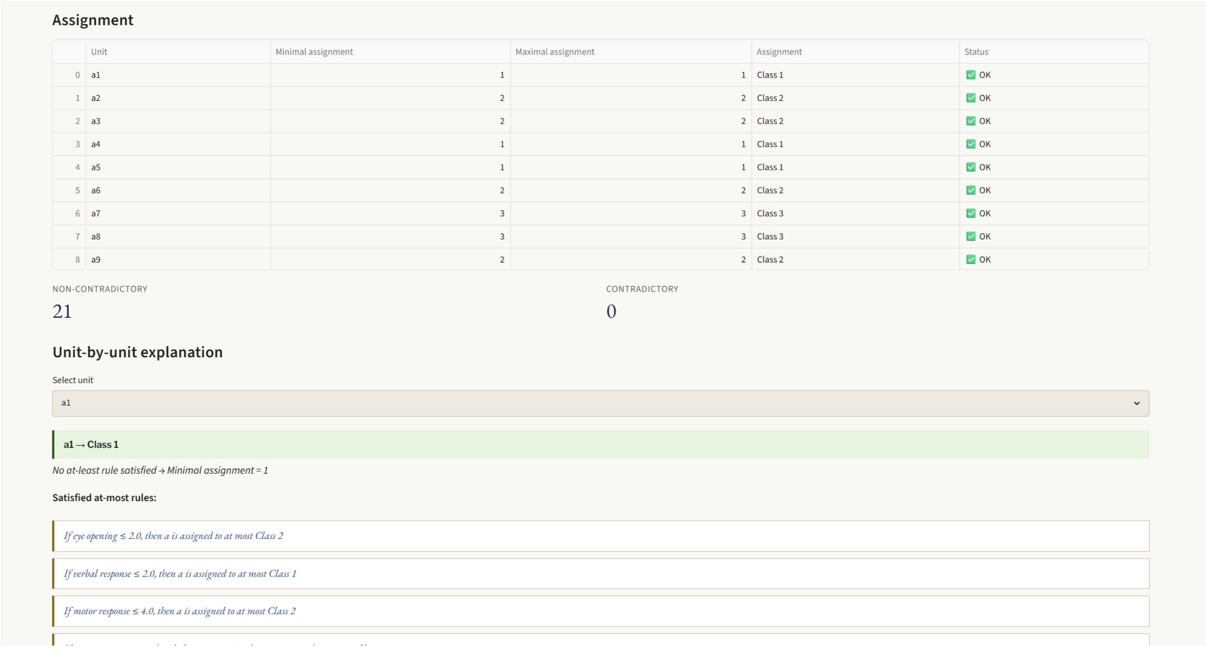

Figure 12: Tab APPLY RULES - Assignment results from uploaded induced rules

8.4 Load a new unit dataset

After loading a set of rules and a set of units with non-contradictory assignments, one may investigate if the introduction of new units affects rules selection and all the units assignments.

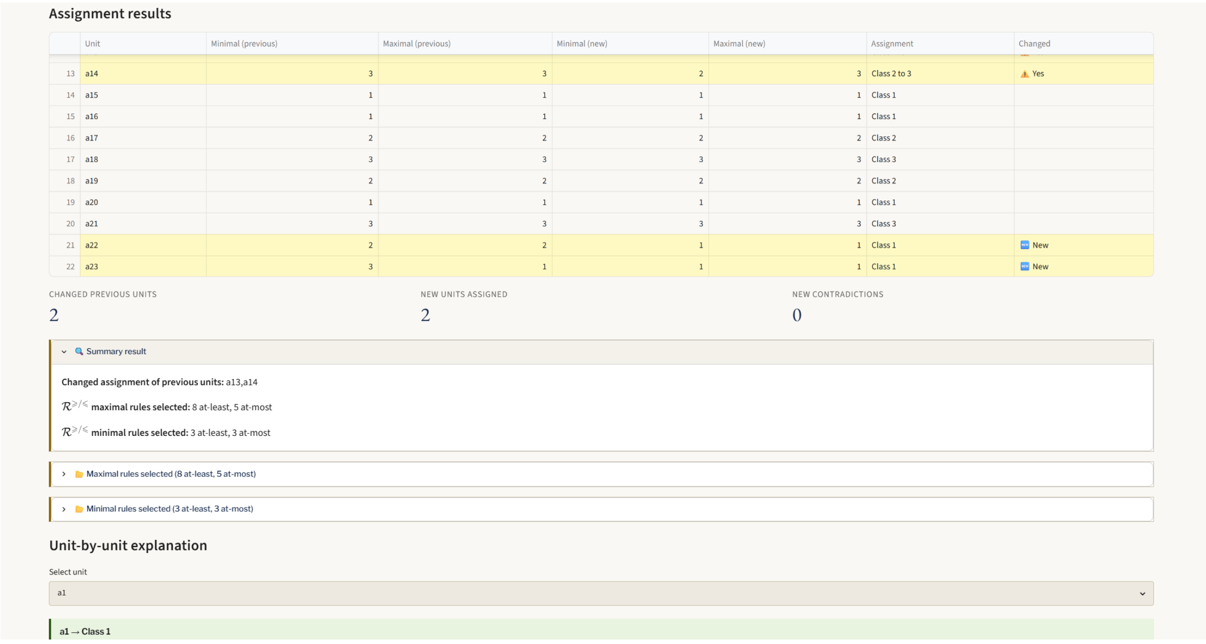

Figure 13: Tab APPLY RULES - Assignment results and rule selection after new units uploaded

The shown displayed information is similar to the one in Section 7.2. Moreover, unit-by-unit explanation, and CSV/EXCEL download are also available.

Tip

The result would be the same as running the full pipeline in the RUN tab, and then running the INCREMENTAL LEARNING tab.

9. Examples

All example files are available in the `examples/` folder of the repository (link).

9.1 Example 1 - Human Development Index

Files: `examples/Ex.1_HDI.csv`.
Problem: Inducing rules + inducing minimal set of rules (all reference units) + classification.
Dataset: 193 countries evaluated on 4 criteria (life expectancy, expected years of schooling, mean years of schooling, GNI per capita), assigned to 4 development classes.
Directions: all increasing (↑).

Workflow:

1. SIDEBAR: upload `Ex.1_HDI.csv` and set the separator `','`.
2. DATA & SETUP: set all criteria as increasing and ensure all units are recognized as reference ones.
3. RUN: set *Classification* and *Rule induction only*. Click on *Run*.
4. RUN: inspect induced rules.
5. RUN: click on *Find minimal set*.
6. RUN: inspect minimal set of rules.
7. ASSIGNMENT: inspect the countries assignment.

9.2 Example 2 - Stock Portfolio

Files: `examples/Ex.2_PTF.txt`, `examples/Ex.2_PTF_new_units.txt`.
Problem: Full pipeline + Incremental learning.
Dataset: 50 stocks evaluated on 8 financial criteria (Return, Beta, Net Profit Margin, ROA, ROE, EPS, P/E, P/BV), assigned to 3 performance classes.
Directions: all increasing (↑).

Workflow:

1. SIDEBAR: upload `Ex.2_PTF.txt` and set `';'` as separator.
2. DATA & SETUP: set all criteria as increasing.
3. RUN: select *Full pipeline* and click on *Run*.
4. ASSIGNMENT: inspect stocks assignment.
5. INCREMENTAL LEARNING: upload `Ex.2_PTF_new_units.txt` and set `';'` as separator. Click on *Assign new units*.
6. INCREMENTAL LEARNING: inspect any changes.

9.3 Example 3 - Scored units

Files: `examples/Ex.3_ElectreScore.txt`.
Problem: Scoring.
Dataset: 10 investment projects evaluated on 5 criteria, ranked by a continuous score from 0 to 100.
Directions: g1 (↓), g2 (↓), g3 (↑), g4 (↑), g5 (↑).

Workflow:

1. SIDEBAR: upload `Ex.3_ElectreScore.txt` and select `'\t'` as separator.
2. DATA & SETUP: set criterion directions (decreasing, decreasing, increasing, increasing, increasing).
3. RUN: select *Scoring* as problem type, select *Rule induction only* and click on *Run*.
4. RUN: inspect rules - note that rules decision part shows original score values.
5. ASSIGNMENT: inspect assignment (shows score values).

9.4 Example 4 - Missing Values

Files: `examples/Ex.4_MISSING.csv`, `examples/Ex.4_MISSING_new_units.csv`.
Problem: Full pipeline + Class assignment with missing values.
Dataset: 20 students evaluated on 3 criteria (mathematics score, reading score, socioeconomic index), assigned to 3 performance classes. Several missing values are present in the evaluation of the units.
Directions: all increasing (↑).

Workflow:

1. SIDEBAR: upload `Ex.4_MISSING.csv` and set `','` as separator.
2. DATA & SETUP: inspect NaN/missing values are correctly recognized.
3. RUN: select `Classification` and *Full pipeline*. Click on *Run*.
4. ASSIGNMENT: inspect class assignment.
5. INCREMENTAL LEARNING: upload `Ex.4_MISSING_new_units.csv` and select `','` as separator. Click on *Assign new units* and inspect.

9.5 Example 5 - Apply Rules (quantitative)

Files: `examples/Ex.5_GCS_original.xlsx`, `examples/Ex.5_GCS_new_units.xlsx`, `examples/Ex.5_maximal_rules.csv`.
Problem: Load previous induced rules + assign units + assign new units.
Dataset: Pre-computed rules for the Glasgow Coma Scale (GCS) dataset (9 at-least, 8 at-most rules, 3 criteria, class mode).

Workflow:

1. APPLY RULES: upload `Ex.5_maximal_rules.csv` and inspect the upload rules.
2. APPLY RULES: upload `Ex.5_GCS_original.xlsx` and inspect their class assignment and unit-by-unit explanation.
3. APPLY RULES: upload `Ex.5_GCS_new_units.xlsx` and click on *Assign new units*.
4. APPLY RULES: inspect new units assignment.

9.6 Example 6 - Classification with qualitative criteria

Files: `examples/Ex.6_GCS_qualitative.xlsx`.
Problem: Induce rules with qualitative criteria.
Dataset: Glasgow Coma Scale (GCS) dataset with qualitative entries.
Directions: all increasing (↑).
Mapping: Eye opening –> Not (1), To pain (2), To sounds (3), Spontaneous (4); Verbal response –> Not (1), Incomprehensible sounds (2), Inappropriate words (3), Confused conversation (4), Oriented (5); Motor response –> Abnormal extension (1), Abnormal flexion (2), Withdraws from pain (3), Localizes pain (4), Obeys commands (5); Severity class –> Severe (1), Moderate (2), Mild (3).

Workflow:

1. SIDEBAR: upload `Ex.6_GCS_qualitative.xlsx`.
2. DATA & SETUP: fill *Qualitative criteria mapping* and *Criteria preference direction*.
3. RUN: select *Classification* as problem type and *Rule induction only* as Mode. Click on *Run* button.
4. RUN: inspect induced rules.
5. RUN: click on *Find minimal set* and inspect selected rules
6. ASSIGNMENT: inspect unit assignments.

9.7 Example 7 - Apply Rules (qualitative)

Files: `examples/Ex.7_qualitative_maximal_rules.csv`, `examples/Ex.7_GCS_qualitative.txt`, `examples/Ex.7_GCS_qualitative_new.xlsx`.
Problem: Load previous induced rules with qualitative criteria + assign units + assign new units.
Dataset: Pre-computed rules for the GCS dataset with qualitative entries (9 at-least, 8 at-most rules, 3 criteria, class mode).

Workflow:

1. APPLY RULES: upload `Ex.7_qualitative_maximal_rules.csv` and inspect the upload rules.
2. APPLY RULES: upload `Ex.7_GCS_qualitative.txt` and select `','` as separator.
3. APPLY RULES: inspect their class assignment and unit-by-unit explanation.
4. APPLY RULES: upload `Ex.7_GCS_qualitative_new.xlsx` and click on *Assign new units*.
5. APPLY RULES: inspect new units assignment.

9.8 Example 8 - Scoring with mixed quantitative and qualitative criteria

Files: `examples/Ex.8_ElectreScore_mixed.xlsx`,
`examples/Ex.8_ElectreScore_mixed_new.xlsx`.
Problem: Full pipeline with quantitative and qualitative criteria + Incremental learning.
Dataset: 10 investment projects evaluated on 5 criteria, ranked by a continuous score from 0 to 100. Criteria with quantitative and qualitative entries.
Directions: Investment cost (↓), Annual cost (↓), Recruitment (↑), Image (↑), Access (↑).
Mapping: Recruitment –> Very bad (1), Bad (2), Rather bad (3), Average (4), Rather good (5), Good (6), Very good (7); Image –> Very bad (1), Bad (2), Rather bad (3), Average (4),

Rather good (5), Good (6), Very good (7); Access –> Very bad (1), Bad (2), Rather bad (3), Average (4), Rather good (5), Good (6), Very good (7).

Workflow:

1. SIDEBAR: upload `Ex.8_ElectreScore_mixed.xlsx`.
2. DATA & SETUP: fill *Qualitative criteria mapping* and *Criteria preference direction*.
3. RUN: select *Scoring* as problem type and *Full pipeline* as Mode. Click on *Run* button.
4. RUN: inspect induced rules.
5. ASSIGNMENT: inspect unit assignments.
6. INCREMENTAL LEARNING: upload `Ex.8_ElectreScore_mixed_new.xlsx` and click on the *Assign new units button*.
7. INCREMENTAL LEARNING: inspect all units assignments and selected maximal and minimal rules.

10. Troubleshooting

10.1 File format errors

- **"Could not read file: . . . "**: the application could not parse the uploaded file. The most common cause is a mismatch between the actual file separator and the separator selected in the dropdown (comma, semicolon, tab, space). Try changing the separator and re-uploading.
- **"Could not parse the file correctly. The dataset must have at least one criterion and one class column. Check the separator setting."**: the file was read but resulted in a single column, meaning the separator is wrong. Change the separator drop-down to match the file format.
- **Criteria columns show unexpected values**: ensure that the class/score label is in the **last** column of the file. All intermediate columns are treated as criteria.
- **Non-numerical values in criteria**: columns with non-numeric values (other than the first column) are automatically recognized as qualitative criteria. The Qualitative criteria mapping widget appears in the DATA & SETUP tab. If you did not intend to use qualitative criteria, check that the file separator is correct and that all criteria columns contain numeric values only.
- **No reference units detected**: if all class/score labels are empty, no rules can be induced. Ensure that at least some units have a numeric entry in the last column. Note: if the class column is qualitative, reference units are those with a non-empty class value.
- **Excel files with multiple sheets**: when uploading an `.xlsx` file, only the first sheet is read. Ensure that the data is in the first sheet.

10.2 Data & Setup errors

- **"Please select at least one reference unit."**: at least one reference unit must be selected in the multiselect widget before proceeding to Tab Run.
- **"Reference units must cover at least 2 distinct classes to induce rules. Please select reference units from at least 2 different classes."**: rule induction requires reference units from at least two distinct classes. If all selected reference units belong to

the same class, no rules can be induced. Add reference units from a second class, or check that the class column in the dataset is correctly populated.

10.3 Qualitative criteria errors

- **Duplicate ranks warning**: if two or more values in the same criterion are assigned the same rank, a warning is displayed. Each value must have a unique rank.
- **"Value(s) [...] of criterion '...' not found in mapping."**: a value in the uploaded units file (Tab INCREMENTAL LEARNING or Tab APPLY RULES) does not appear in the qualitative mapping. This can happen if the units file contains a value that was not present in the original dataset or rules file. Check that all qualitative values in the units file are listed in the mapping.
- **Default alphabetical ordering**: if no mapping is specified by the user, qualitative values are ranked in alphabetical order (A=1, B=2, . . .). This may not reflect the correct preference order. Always verify and adjust the mapping before running the analysis.
- **Class column qualitative**: if the last column (class) contains qualitative values, it is also included in the mapping widget. Ensure the ranking reflects the correct preference order (1 = worst class, p = best class).
- **Mixed criteria**: datasets with both numeric and qualitative criteria are supported. Only non-numeric columns are shown in the mapping widget; numeric criteria are used as-is.

10.4 Rules file errors (Tab APPLY RULES)

- **"Missing `#directions` line in rules file. The rules file must start with a line like: `#directions,increasing,decreasing,...`"**: the uploaded rules file does not contain the required metadata header. Check the file format described in Section 3.
- **"Missing `#mode` line in rules file. The rules file must contain a line like: `#mode,class` or `#mode,score`"**: the rules file is missing the mode declaration. Add a `#mode,class` or `#mode,score` line after the `#directions` line.
- **"The rules file contains no rules."**: the rules file was parsed correctly but contains no rule rows. Check that the file has data rows below the header.
- **"The uploaded units file has no criteria columns matching the rules. Expected: . . . Please upload a file with the same criteria names."**: none of the column names in the units file match the criterion names declared in the rules file. Column names are case-sensitive and must match exactly. Extra columns in the units file are silently ignored.
- **"Missing criteria: . . . These will be treated as NaN."**: some criteria declared in the rules file are absent from the units file. The missing criteria will be treated as missing values. If this is unintended, check the column names.
- **"The uploaded units file is empty."**: the units file contains no data rows. Upload a non-empty file.
- **"The uploaded units file contains no valid numeric values."**: all values in the units file are non-numeric or missing after parsing. Check that the file uses the correct separator and that criteria columns contain numeric data.
- **Wrong `#mode`**: if rules were exported from a scoring problem, the file contains `#mode,score`. Loading it in Tab Apply Rules will display score values instead of class indices. This is correct behavior.

10.5 New units errors (Tabs INCREMENTAL LEARNING and APPLY RULES)

- **"The uploaded file has no criteria columns matching the original dataset. Expected: ... Please upload a file with the same criteria names."** (Tab INCREMENTAL LEARNING): none of the column names in the new units file match the criterion names of the original dataset. Column names must match exactly (case-sensitive). Extra columns are ignored.
- **"Missing criteria columns: ... These will be treated as NaN."**: some criteria are absent from the new units file and will be treated as missing values.
- **"The uploaded file contains no valid numeric values."**: the new units file contains no usable numeric data. Check the separator and column contents.
- **"The uploaded file is empty."**: the new units file has no data rows.
- **"Could not assign new units: ... Make sure the new units file has the same criteria as the original dataset."**: a dimension mismatch occurred during the assignment computation. This typically happens when the new units file has a different number of criteria than the original dataset.
- **"Unexpected error: ... "**: an unhandled error occurred. Try reloading the page and repeating the operation. If the problem persists, check the input files for anomalies.

10.6 Pipeline errors

- **"Run the Full pipeline first"** (Tab INCREMENTAL LEARNING): the New Units tab requires the full pipeline to have been completed in Tab Run. Go to Tab Run, select *Full pipeline*, and click **Run**.
- **"No rules could be induced. Check that reference units cover at least 2 classes and the confidence level is not too high."**: rule induction returned an empty rule set. This can happen if the confidence level slider is set too high, or if the reference units do not provide sufficient information to distinguish between classes. Try lowering the confidence level or reviewing the reference unit selection.
- **"MILP minimisation failed (...). Using maximal rules (... at-least, ... at-most) instead."**: the MILP solver could not find a minimal set of rules within the time or feasibility constraints. The application automatically falls back to the maximal rule set. Results remain valid; only the minimality guarantee is lost.
- **Rule induction takes a long time**: for large datasets or datasets with many contradictions, rule induction and the MILP step may take several minutes. This is expected behavior.

How to Cite

If you use this software in your research, projects, or publications, please cite the following papers.

- Corrente, S., Greco, S., Słowiński, R., & Zappalà, S. (2026). *An explainable and interpretable composite indicator based on decision rules*. Omega, 142, 103513. `https://doi.org/10.1016/j.omega.2026.103513`.
- Greco, S., Matarazzo, B., & Słowiński, R. (2001). *Rough sets theory for multicriteria decision analysis*. European Journal of Operational Research, 129(1), 1-47. `https://doi.org/10.1016/S0377-2217(00)00167-3`.

For BibTeX users you can cite them directly in your LaTeX document using `\cite{corrente2026}` and `\cite{greco2001}`:

```
@article {corrente2026,
title = { An explainable and interpretable composite indicator
based on decision rules },
author = { Corrente, Salvatore and Greco, Salvatore and S{\l}owi\'{n}
   ski, Roman and Zappal\`{a}, Silvano },
journal = { Omega },
volume = {142},
pages = {103513},
year = {2026},
publisher = { Elsevier },
doi = {10.1016/j.omega.2026.103513}
}
@article{greco2001,
title = { Rough sets theory for multicriteria decision analysis
},
author = {Greco, Salvatore and Matarazzo, Benedetto and
S{\l}owi\'{n}ski, Roman},
journal = { European Journal of Operational Research },
volume = {129},
number = {1},
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year = {2001},
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doi = {10.1016/S0377-2217(00)00167-3}
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